

Project Requirement Document

1. Project Overview

Objective:

Build an automated system that helps secure visa appointment slots on **vfsglobal.com** for:

- **Angola → Brazil**
- **Angola → Portugal**

The system should continuously monitor appointment availability, auto-fill applicant details, and book appointments as soon as slots open, with manual and automated control options.

This document is intended for **developers**, so requirements are written in technical and implementation-friendly language.

2. Problem Statement

VFS Global appointments for Angola are:

- Released **irregularly**
- Booked within **seconds**
- Often blocked by captchas, session limits, and IP rate limits

Manual booking is inefficient and unreliable. We need a **smart, resilient automation solution**.

3. Scope of Work

In-Scope

- Appointment monitoring
- Auto-form filling
- Slot booking
- Multi-profile handling
- Notification system
- Admin dashboard

Out of Scope (Phase 1)

- Payment automation (only manual payment support)
 - Passport delivery tracking
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4. Target Platforms

- **Website:** <https://visa.vfsglobal.com/>
 - **Countries:**
 - Origin: Angola
 - Destination: Brazil, Portugal
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5. Functional Requirements

5.1 Appointment Monitoring

- Select:
 - Country of residence (Angola)
 - Destination country (Brazil / Portugal)
 - Visa category (Tourist, Business, etc.)
 - Auto-refresh appointment availability
 - Configurable refresh interval (e.g. 5s–60s)
 - Detect newly released slots in real time
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5.2 Auto Booking System

- Auto-select earliest available date & time
 - Auto-submit booking form instantly
 - Support **manual override** before final submission
 - Retry mechanism if booking fails
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5.3 Applicant Profile Management

- Save multiple applicant profiles:
 - Full name
 - Passport number

- DOB
 - Passport expiry
 - Nationality
 - Email & phone
 - Profile tagging (High priority / Normal)
 - One-click profile selection
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5.4 Captcha Handling Strategy

- Detect captcha type (reCAPTCHA / image captcha)
 - Options:
 - Manual captcha input popup
 - Captcha-solving service integration (2Captcha / Anti-Captcha – optional)
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5.5 Proxy & IP Rotation

- Residential / mobile proxy support
 - Country-specific IP routing
 - Auto-IP rotation on block detection
 - Session persistence via cookies
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5.6 Notifications

- Instant alerts when:
 - Slot detected
 - Appointment booked
 - Booking failed
 - Notification channels:
 - Telegram Bot
 - Email
 - Desktop notification
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5.7 Logs & Reports

- Timestamped logs:
 - Slot detection
 - Booking attempts

- Errors & blocks
 - Export logs (CSV / TXT)
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6. Non-Functional Requirements

- High reliability & uptime
 - Low latency during slot release
 - Scalable for 50–200 profiles
 - Human-like behavior simulation
 - Secure storage of user data
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7. Suggested Tech Stack

7.1 Backend / Automation

- **Language:** Node.js or Python
 - **Automation:**
 - Playwright (preferred)
 - Puppeteer (alternative)
 - **Captcha:** 2Captcha / Anti-Captcha API
 - **Proxies:** BrightData / SmartProxy / Oxylabs
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7.2 Frontend (Dashboard)

- React.js / Next.js
 - Tailwind CSS or Material UI
 - Responsive (Desktop-first)
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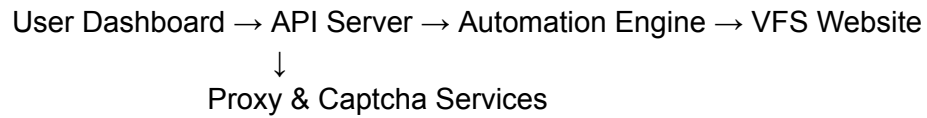
7.3 Database

- PostgreSQL / MongoDB
 - Encrypted sensitive fields
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7.4 Notifications

- Telegram Bot API
 - SMTP (Gmail / SendGrid)
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8. System Architecture (High Level)



9. UI / UX Design Requirements

9.1 Login Screen

- Email / password
 - Role-based access (Admin / Operator)
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9.2 Dashboard

- Appointment status cards
 - Live logs panel
 - Active sessions counter
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9.3 Appointment Setup Screen

- Dropdowns:
 - Origin country
 - Destination country
 - Visa type
 - Date range preference
 - Auto / Manual mode toggle
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9.4 Applicant Profiles Screen

- Add / edit / delete profiles

- Bulk upload via Excel
 - Priority flag
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9.5 Logs & History Screen

- Filter by date / profile / result
 - Download logs
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10. Security Considerations

- Encrypt stored credentials
 - Avoid hardcoded secrets
 - Rate limiting on dashboard actions
 - Proxy anonymization
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11. Success Criteria

- Appointment booked within **seconds** of slot release
 - Minimal bans / blocks
 - Easy operation by non-technical users
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